



ProofyX

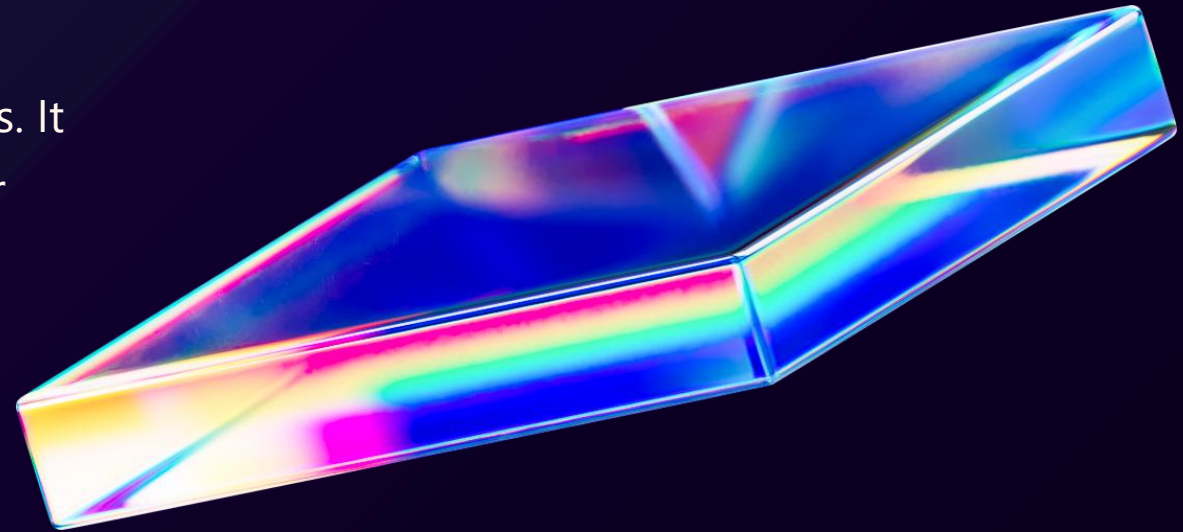
Is this real?

Explainable media forensics for fraud teams and investigators. It tells you whether an image, video or voice is AI-generated or tampered — and shows you the evidence.

Built at B.S. Abdur Rahman Crescent Institute of Science & Technology

CIIC incubation application

₹8 lakh · 6 months · 5 outcomes



In India, the deepfake is now a fraud channel.

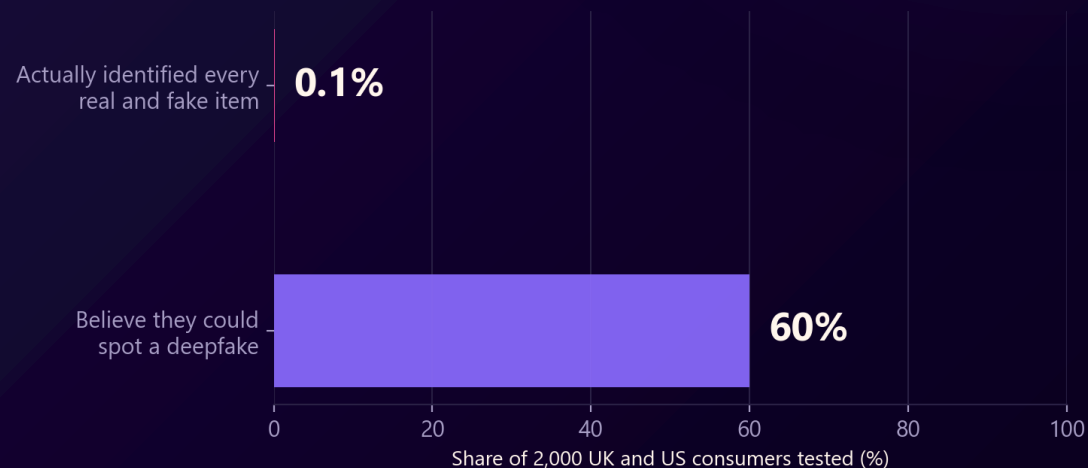
People cannot tell what is real — and regulators are now asking them to.

₹22,495 cr Lost to cyber fraud in India in 2025 —
28.15 lakh complaints, up 24% (I4C / MHA)

47% Of Indians have been a victim of, or know a
victim of, an AI voice-cloning scam (McAfee)

**1 Sept
2026** TN Cyber Crime Wing registers a CBCID case
over a deepfake video of the Chief Minister

65% of Indian organisations have already experienced a deepfake-driven attack
(Thales, 2026).



iProov tested 2,000 UK and US consumers on real and fake media. Six in ten backed themselves; one in a thousand got every item right.

Detection is now a compliance duty.

The buyer no longer needs convincing that the problem exists. A regulator told them.

MEITY · IT RULES

IT Amendment Rules 2026

In force 20 Feb 2026.
Synthetically generated information must be clearly labelled; platforms carry verification and takedown duties.

RBI · KYC

KYC amendment, 14 Aug 2025

Video-KYC (V-CIP) must actively detect AI-generated faces, replays and masks — basic liveness prompts are no longer enough.

MHA · I4C

27.3 lakh mule accounts flagged

Layer-1 mule accounts identified by January 2026. Every one passed a KYC check. RBI's MuleHunter.AI is live across 26 banks.

EXPORT TAILWIND

EU AI Act, Article 50

From 2 Aug 2026 AI-generated content must be machine-detectable — the same capability, sold abroad later.

Banks, NBFCs and investigators now have to answer 'is this real?' as a matter of compliance — and most of them have no tool for it.

One score, and the reason behind it.

Image, video or audio in. A calibrated trust score, a heat-map and a per-model breakdown out — the thing a compliance officer can put in a file.



Cross-modal fusion: a video is scored on its frames AND its audio track, and disagreement between modalities is surfaced, not hidden.

EVIDENCE, NOT A VERDICT

Grad-CAM shows where; every model reports separately.

INDIA-FIRST WORKFLOWS

Aadhaar / PAN / Voter-ID checks,
cybercrime.gov.in-format complaint document,
IT-Rules traceability API.

DEPLOYABLE

Runs on CPU; on-prem path for banks and
police who cannot send evidence to a foreign
cloud.

Upload. Score. Explain.

Live build, laptop CPU, no GPU — about 5–6 s per image end to end. One flow: a known fake in, evidence out.

1

Upload

Drag in a file, or POST to /api/v1. JPG PNG WebP · MP4 MOV MKV · WAV MP3 FLAC.

2

Analyse

The model bank scores in parallel; learned fusion reconciles them.

3

Explain

Trust score 0–100, Grad-CAM heat-map, per-model scores, EXIF report.

4

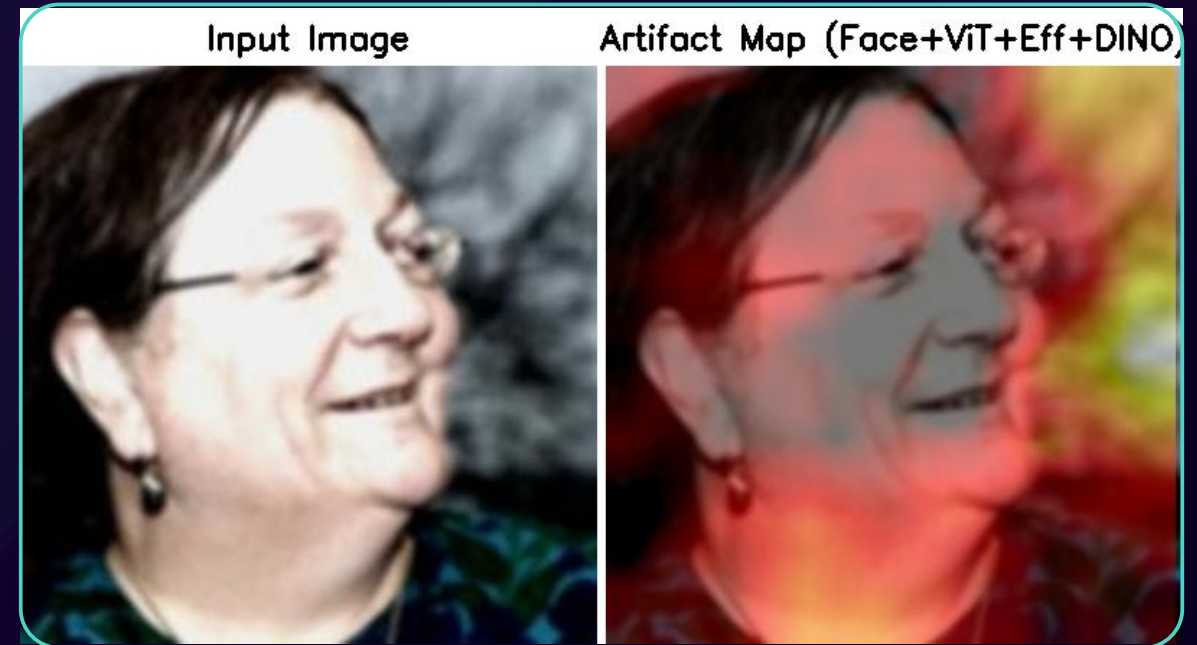
Act

Verdict + confidence band, and a cybercrime.gov.in-format complaint document.

AI-GENERATED ≥ 0.60 AUTHENTIC < 0.60

CONFIDENCE HIGH / MED / LOW

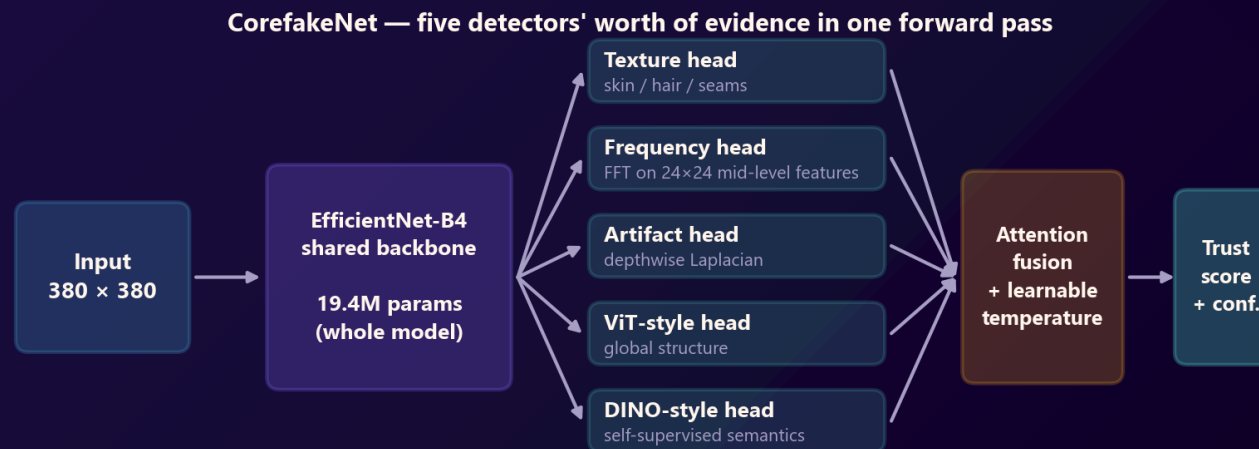
Verdict plus a separate confidence band, so a 0.55 never reads like a 0.95.



Real output from the current build — a known deepfake portrait, ensemble verdict AI-GENERATED at risk 0.65, 3/7 models detect manipulation, 25.0 s on a laptop CPU.

One backbone, five forensic heads.

A seven-model ensemble is accurate but slow. We trained a single multi-head network to replace it — and we intend to patent the fusion design.



Multi-task loss: per-head BCE + fusion BCE + KL diversity + confidence regulariser. Measured: 564 ms/image on CPU, 4.9× faster than the 7-model ensemble.

19.4M

parameters, whole model

564 ms

model forward on CPU;
~5–6 s end to end

4.9×

vs the 7-model ensemble
(2,759 → 564 ms forward)

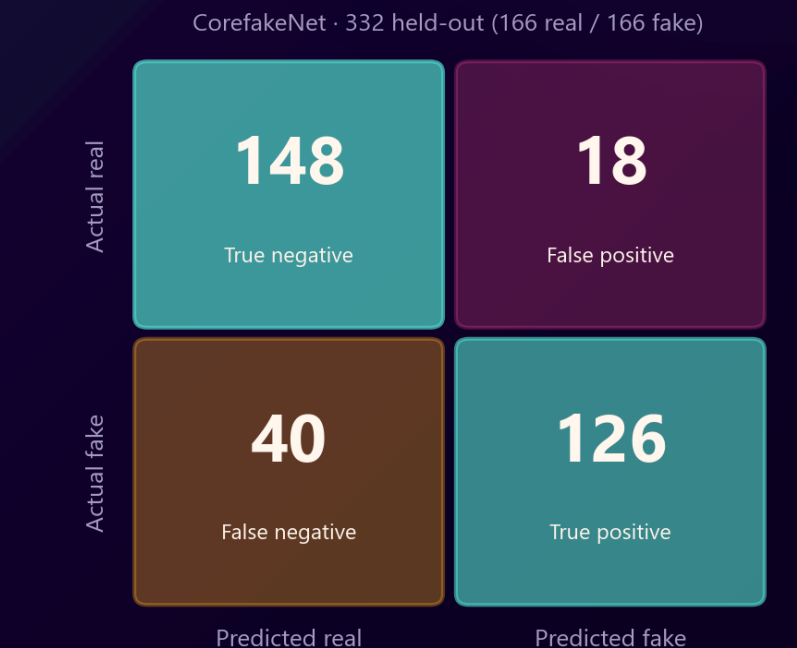
Month 2

provisional patent on the fusion
design, via CLIC's IP cell

What is measured, and what is not.

We label targets as targets. The amber and magenta rows are what ₹8 lakh is for.

Capability	Status	Result	Scope
CorefakeNet image model	Measured	82.5% acc · 90.9% ROC-AUC · F1 81.3	332 held-out
Latency	Measured	564 ms model forward on CPU; ~5–6 s end to end	Benchmark run
Video pipeline	Verified	End to end at 1 / 2 / 10 / 20 fps	Live build
Audio (Wav2Vec2)	Upstream	97.9% is the model card, not ours	Own eval month 3
7-model ensemble	TARGET	95%+ — calibration bug found; fix month 1	Not claimable
In-domain spot check	Measured	62% on 40 portrait samples at the 0.60 threshold (7/20 fakes caught, 2 false alarms)	This week, fast mode
Out-of-domain (OpenFake)	GAP	46% on 24 illustration/scene samples — generalisation is the gap	Benchmark month 3
Adversarial robustness	TARGET	JPEG / resize / blur sweep	Month 2
Paying customers	GAP	Zero — design partners month 5	Honest baseline



Reconstructed from the reported precision / recall / accuracy. The committed eval output replaces this in month 1.



Start where the **budget already exists.**

Not consumers. Three partners in six months — one of them paying.

PAID PILOT

NBFC / fintech video-KYC fraud teams

RBI's Aug 2025 amendment made deepfake detection in V-CIP their obligation. They pay per check, and Chennai is dense with them.

VALIDATION PILOT

A Tamil Nadu cyber-crime / forensics unit

Unpaid, through CIIC's government network. This is what gives us credibility with everyone else.

DESIGN PARTNER

One newsroom

Easiest access, weakest payer — kept for workflow feedback, not revenue.

What a pilot is

8 weeks · one workflow · a success metric agreed up front · hosted or on-prem · pricing hypothesis ₹2–10 per check, tested in the pilot, not assumed.



A funded field, an India-shaped gap.

We know who is here. We are not claiming the best accuracy — we are claiming the most auditable, deployable answer for regulated mid-market teams.

28.15 lakh

cyber-crime complaints in India, 2025
(the demand side of forensics)

Every V-CIP

RBI-regulated bank, NBFC and fintech doing
video-KYC now needs deepfake checks

\$170M → \$5.6B

global detection market 2025 → 2034,
47.6% CAGR (Market.us)

Player	Base	Focus	Explainability	On-prem / CPU
pi-labs	Pune	LEA, defence, BFSI	Forensic reports	Yes
Kroop AI	India	KYC, enterprise	Not published	Not published
Contrails AI	India	Platforms, govt	Not published	Not published
Reality Defender	US	Multi-modal	Partial	No
Pindrop	US	Voice	No	—
ProofyX	Crescent	Image + video + audio	Grad-CAM + per-model	CPU · on-prem

pi-labs is the incumbent with Indian law enforcement and is in Gartner's 2026 deepfake-detection quadrant. Our claim is narrower: explainable, India-first, on-prem CPU — for the regulated mid-market the big vendors do not serve.

Four ways to charge for one API call.

Land through per-check and team plans; expand into on-prem licences. Every price below is a hypothesis until a pilot tests it.

HYPOTHESIS — tested in pilots

Stream	Price (hypothesis)	Who	Why they buy
Per-check API	₹2–10 per check	NBFC / fintech KYC vendors	Regulatory duty, priced per user
Team plan	₹8,000–40,000 / month	Newsrooms, SMB fraud desks	Desk-level verification
On-prem licence	₹5–50 lakh / year	Banks, government	Data residency, SLA, audit trail
Government project	Project pricing	Cyber-crime units, forensic labs	Evidence workflow, training

UNIT COST

~5–6 s per image end to end on a laptop CPU today; the model forward is 0.56 s. Compute per check is paise either way — the cost is sales, not inference.

LAND AND EXPAND

One integration serves KYC, moderation and evidence — three budget lines in the same customer.

DATA RESIDENCY

On-prem and Indian-cloud deployment is the door-opener for banks and police; foreign APIs cannot follow.



What CIIC has is what we need next.

Not flattery — a map of CIIC assets to ProofyX milestones. Every line is from CIIC's own pages.

MISSION

CIIC exists to turn Crescent students into startups.

We are Crescent students with a running, measured product.

DOMAIN

AI/ML and Cyber Security are on CIIC's Industry 4.0 list.

ProofyX is an AI-security product — not media, not consumer.

COMPUTE

Intel AI Lab and Cloud Computing Lab.

The difference between 'target 95%' and a measured number.

IP

CIIC is the university's nodal IP office (CIPR / CTT).

Provisional patent on the fusion design, month 2.

FUNDING PATH

Implementing agency for MeitY GENESIS EiR and SAMRIDH; DST NIDHI-PRAYAS centre.

We ask to be hosted for the next open window.

NETWORK

IHFC (IIT Delhi) CiC; IITMIC co-incubation MoU; government and BFSI ties.

Technical depth we cannot get alone; our first pilots.

Ask: incubation now · hosting for the next scheme window · a named mentor from CSE.

₹8 lakh: prototype to validated pilots.

Six months, five outcomes. Not CIIC's balance sheet — a CIIC-hosted scheme window plus in-kind support.

₹ lakh	Activity	Milestone
1.60	GPU / cloud compute — calibration retraining, robustness sweep, pilot inference	M2 measured ensemble
0.90	Benchmark & evaluation data, annotation, harness	M3 public benchmark
0.60	Security hardening + external VAPT of the API	M4 VAPT report
0.90	Production API, on-prem packaging, pilot environment	M4 pilot live
1.20	Incorporation, DPIIT, trademark, provisional patent, pilot MoU templates	M1 entity · M2 filing
0.60	Customer validation & pilots — travel, demos, workshops	M5 three partners
1.40	Founder sustenance / part-time technical help (≈₹23k per month, team)	Full-time focus
0.80	Contingency (10%)	—
8.00		

WHERE IT COMES FROM

- CIIC-hosted grant: NIDHI-PRAYAS 2.0 (≤₹20 lakh), GENESIS EIR (≤₹10 lakh) or MSME Innovative (≤₹15 lakh) — whichever opens first
- In kind: Intel AI Lab compute, IP filing, DPIIT and legal support
- Crescent Seed Money (₹20,000 per student) as a bridge

FIVE OUTCOMES BY MONTH 6

- 1 Ensemble calibration fixed and measured
- 2 A public-benchmark result anyone can reproduce
- 3 A security-audited, pilot-ready API
- 4 Three design partners under LOI
- 5 One paid pilot running

M0

Incorporate · DPIIT · onboard

M1

Calibration fixed

M2

Robustness · patent filed

M3

Public benchmark

M4

VAPT · pilot env live

M5

3 design partners

M6

1 paid pilot · Stage-2 plan

Then ₹20–30 lakh. Then ₹50 lakh to ₹1 crore.

₹8 lakh is the first milestone-based requirement, not the lifetime need. Each stage is unlocked by evidence from the last.



Muhammed Sayeedur Rahman
S

ML & platform · detection stack, fusion,
API, compliance layer

Mohamed Abrar

Product & frontend · dashboard,
analysis workflows, deployment

Mohamed Shahid

Infrastructure & QA · CI/CD, containers,
security scanning, tests

WE ARE ASKING CIIC FOR

- Incubation-committee slot
- Named CSE faculty mentor
- Scheme-application conversation

Full-time commitment and department / year for each founder to be stated in the room. Roles above are from repository ownership — confirm before presenting.



Shipped, experimental, planned.

For questions. Repository at HEAD, September 2026 — 251 automated tests.

Capability	Status	Note
Image detection — ensemble (DINOv2, EfficientNet, ViT, CLIP, face, texture, frequency)	SHIPPED	Ensemble accuracy not yet measured (calibration bug)
CorefakeNet unified model	SHIPPED · MEASURED	82.5% / 90.9% AUC / 564 ms
Grad-CAM + per-model evidence	SHIPPED	
Video (frame sampling, timeline, temporal LSTM)	SHIPPED	LSTM unvalidated
Audio (Wav2Vec2 wrapper + CNN fallback)	SHIPPED	Upstream model; own eval month 3
Multimodal fusion	SHIPPED	Flagged / clean modalities surfaced
EXIF, GPS, C2PA check, SHA-256 hash	SHIPPED	
Compliance & traceability API (orgs, keys, ingest, SLA, audit log, webhooks)	SHIPPED	IT Rules 2026 alignment
Cyber-crime complaint document	SHIPPED	Document only — no custody chain
Document / ID forensics (ELA etc.)	EXPERIMENTAL	Classical CV heuristics
Aadhaar / PAN / Voter-ID validators	SHIPPED	Format & checksum only
Reverse image search	PARTIAL	Needs Bing key
Adversarial robustness eval	SCRIPT ONLY	Run = month 2
Public benchmark · GPU inference	PLANNED	Months 3 · Stage 2



Every number has a source.

Full research and links: docs/pitch/CIIC_STRATEGY.md

INDIA

- ₹22,495 cr, 28.15 lakh complaints — I4C / MHA, 2025
- 47% AI voice-scam exposure — McAfee, 2023
- 27.3 lakh mule accounts — I4C, Jan 2026
- CM deepfake case — TN Cyber Crime Wing / CBCID, 1 Sept 2026
- 65% of orgs attacked — Thales, 2026

REGULATION

- IT Amendment Rules 2026 — MeitY, in force 20 Feb 2026
- KYC Master Direction amendment — RBI, 14 Aug 2025
- EU AI Act Art. 50 — 2 Aug 2026

CIIC

- Vision, mission, domains — ciic.ventures/about, /nidhi-prayas-program
- GENESIS EiR via CIIC — CIIC X, Jan & Mar 2025
- IHFC CiC — ihfc.co.in, 6 Mar 2025
- IITMIC MoU — crescent.education, 31 Mar 2021
- PRAYAS 2.0 guidelines — nidhi-prayas.org, 2026

MARKET & COMPETITION

- \$170M → \$5.6B, 47.6% CAGR — Market.us
- pi-labs — pi-labs.ai; Gartner 2026
- Kroop AI, Contrails AI — PitchBook, press
- iProov confidence gap — Feb 2025

ProofyX figures: CorefakeNet evaluation on 332 held-out samples; charts regenerated by scripts/make_pitch_figures.py and scripts/make_ciic_figures.py.